Algebraic leak and thermal hybridization, not athermality, defeat quantum many-body scar codes under generic measurement

Hikaru Wakaura and Taiki Tanimae

QIRI (Quantum Integrated Research Institute Inc.), Tokyo 107-0061, Japan

Monitored quantum many-body systems protect quantum information exactly when their unitary dynamics scramble it, placing the measurement-induced transition on the same footing as quantum error correction. Because the eigenstate thermalization hypothesis endows chaotic eigenstates with approximate Knill–Laflamme structure, one asks whether quantum many-body scars—non-thermal eigenstates that stay weakly entangled and dynamically coherent—form good codes. Whether they protect information better or worse than the surrounding thermal states has not been settled. We show that what defeats a scar code is not athermality but two structural liabilities: the coherent information of a logical qubit encoded at the canonical PXP scar rungs lies below that of an energy-matched thermal encoding under every generic measurement we test; only a Casimir probe matched to the scars' multiplet structure reverses the order. The collective deficit is driven by the logical leak of the emergent $\mathrm{su}(2)$ order parameter coupling the code states; the local deficit tracks hybridization with the thermal bulk—protection rises monotonically with FSA purity, crossing to a scar advantage at the purest rungs—while coarse-grained Casimir outcomes are set by bin geometry, not purity. An exactly solvable spin-1 model anchors the zero-variance limit: only zero-variance *exact* scars are protected, their local advantage decaying with rate (size scaling open). The ETH–error-correction correspondence thus holds operationally in the monitored setting, and hybridized scarring—the physically accessed kind—does not aid protection against generic measurement. Structure with respect to the measurement, not athermality, decides the fate of weak-ergodicity-breaking codes; the scar subspaces experiments reach sit at maximal hybridization and are liabilities, not assets, for measurement-based quantum memories.

Hikaru Wakaura: h.wakaura@qiri.co.jp
Taiki Tanimae: t.tanimae@qiri.co.jp

1 Introduction

The competition between scrambling unitary dynamics and local measurements drives a sharp transition in how much quantum information a many-body system retains [1–7]. This measurement-induced phase transition (MIPT) is now understood as a coding transition: below a critical measurement rate the dynamics act as an emergent quantum error-correcting code that hides logical information non-locally, protecting it from measurements that act as erasure errors [8–10]. The order parameter is information-theoretic—the coherent information, or equivalently the entanglement of a reference qubit maximally entangled with the encoded information [9, 11, 12].

The coding power of chaotic dynamics is inherited by chaotic *eigenstates*. The eigenstate thermalization hypothesis (ETH) forces local operators to have smooth diagonal and exponentially small off-diagonal matrix elements [13–16], which is exactly the approximate Knill–Laflamme condition [17]; thermal eigenstates therefore form approximate codes whose error scales with the system's chaoticity [18–20]. Quantum many-body scars are the conspicuous exception to ETH: a measure-zero tower of atypically weakly entangled, dynamically reviving eigenstates embedded in an otherwise thermal spectrum [21–25]. Scars can be embedded in decoherence-free subspaces [26–28], and projective measurement can even enhance their revivals [29], suggesting they might be privileged carriers of quantum information under monitoring.

However, whether a logical qubit stored in the scar subspace is protected better or worse than one stored in the surrounding thermal bulk—at matched energy, in the same monitored dynamics, quantified by the coherent information the monitored channel retains—has not been determined. In this study, we compute the reference-qubit coherent information $\mathcal{C}_R$ of scar and thermal codes of the monitored PXP chain across local, collective, and Casimir measurement models. We find that thermal codes protect information better than the canonical scar code under every generic measurement, with the su(2)-adapted Casimir the encoding-dependent exception; that an apparent scar advantage seen against a single thermal reference is an artifact of an atypical baseline and vanishes under ensemble averaging; that the scar's emergent-su(2) structure is a liability, not a shield, because its generator couples the code states and the scars' excess Casimir variance defeats even a fine-tuned decoherence-free encoding; and—resolving the mechanism—that protection under generic local measurement rises monotonically with the rung's FSA purity, identifying hybridization with the thermal bulk, not athermality, as the controlling liability.

The result establishes the ETH-to-error-correction correspondence as an operational statement about monitored circuits and removes the physically realized, hybridized scar subspaces from the menu of candidate measurement-robust quantum memories.

2 Model and diagnostics

2.1 Model, encoding, and monitored dynamics

We study the PXP chain of L sites in the Rydberg-blockaded (constrained) Hilbert space of dimension $F(L{+}2)$ (Fibonacci), with

$$H = \sum_i P_{i-1}\, X_i\, P_{i+1}, \qquad P = |0\rangle\langle 0|, \tag{1}$$

whose $\mathbb{Z}_2$ Néel state exhibits the canonical scar revivals [21, 22, 30, 31] (period $t \simeq 4.7$, fidelity $\simeq 0.74$ at $L = 12$). We identify the $L{+}1$ scar-tower eigenstates by their anomalous overlap with $|\mathbb{Z}_2\rangle$.

A logical qubit is encoded in a two-dimensional subspace $\{|0_L\rangle, |1_L\rangle\}$ and maximally entangled with a reference R, $|\Psi_0\rangle = (|0\rangle_R |0_L\rangle + |1\rangle_R |1_L\rangle)/\sqrt{2}$. The system evolves by alternating steps of PXP evolution $e^{-iH\,dt}$ and random projective measurement; per quantum trajectory the joint state stays pure and the reference entanglement entropy $S_R = -\mathrm{Tr}\,\rho_R \log_2 \rho_R \in [0, 1]$ measures the surviving logical information. Its trajectory average

$$\mathcal{C}_R(p) = \langle S_R \rangle_{\text{traj}} \tag{2}$$

is the single-qubit coherent information of the monitored channel at fixed circuit depth: $\mathcal{C}_R \to 1$ recoverable, $\mathcal{C}_R \to 0$ lost. Concretely we run 40 evolution–measurement steps and estimate $\mathcal{C}_R$ as the time average of S_R over the last quarter of the record ($t \in [21.6, 24]$). In a finite system at $p > 0$ every code eventually purifies, so $\mathcal{C}_R$ is a fixed-depth retained information, not an asymptotic capacity; what we compare is the scar–thermal *ordering* at matched depth, which is depth-robust wherever information survives the added depth—doubling and quadrupling the depth leaves the collective deficit essentially unchanged (-0.27, -0.34, -0.33 at depths 40, 80, 160; $L{=}14$, $p{=}0.10$) and the local deficit likewise at a rate below the purification scale (-0.14, -0.15, -0.13 at $p{=}0.02$), while at $p = 0.10$ the local channel purifies both codes beyond depth 80 (Appendix B). We take $dt = 0.6$ and measurement probability p per site (local) or per operator (collective) per step. To control the energy-mismatch confound, thermal codes are drawn from generic eigenstates energy-matched to the two scar-rung energies within a tolerance ± 0.5 (realized offsets at $L{=}14$: 0.22 on average, 0.47 at most); **the thermal baseline is averaged over an ensemble of** $\sim$10–20 **such pairs**, without which a single atypical pair inverts the conclusion (Sec. 3). The comparison is insensitive to this tolerance: tightening the window to ± 0.1 leaves the scar deficit unchanged, and within the ensemble $\mathcal{C}_R$ is uncorrelated with the residual offset (Appendix C).

We study $L = 10$–18 (constrained dimension up to $F(20) = 6765$) with 40 evolution–measurement steps per trajectory and average $\mathcal{C}_R$ over 120–400 Born-sampled quantum trajectories. Reported statistical error bars are the standard error of the mean over trajectories; the thermal band is the standard deviation of $\mathcal{C}_R$ over the ensemble of energy-matched thermal pairs. All runs use fixed random seeds and are reproducible from the accompanying code.

Statically we probe the Knill–Laflamme violation of a code under an error set $\{E_a\}$ through the logical leak $\langle 0_L|E_a|1_L\rangle$ and the distinguishability $\langle 0_L|E_a|0_L\rangle - \langle 1_L|E_a|1_L\rangle$.

2.2 A model-independent criterion

A projective measurement of a Hermitian operator O leaves the encoded qubit literally untouched only if the code is an eigenspace of O with a single shared eigenvalue; more generally, the qubit survives a single shot if and only if the Knill–Laflamme conditions hold for the projector set $\{\Pi_g\}$ of O's eigenspaces [17]. A measurement *resolves*—and hence degrades—the code through two channels: *(i) eigenvalue distinguishability*, when the two logical states' outcome distributions $P_a(g) = \langle a_L|\Pi_g|a_L\rangle$ differ; and *(ii) non-invariance*, when a sector-restricted overlap $\langle 0_L|\Pi_g|1_L\rangle$ is nonzero, coherently mixing the logical states. Equal means and a vanishing global leak $\langle 0_L|O|1_L\rangle$ are therefore insufficient: a code spread over many O-sectors—a large O-variance—is collapsed by the shot, and survival then requires the sector-wise conditions in *every* occupied sector, which is why a true decoherence-free subspace demands logical states that are O-eigenstates sharing one eigenvalue. This criterion is independent of the microscopic model and organizes the results below: thermal states, spread broadly and near-identically in a local basis, are close to optimal codes, while scars are penalized whenever the measured operator resolves their atypical structure. Both channels combine into a single exact, simulation-free scalar—the reference entropy $s_1(O)$ surviving one O-measurement, which equals unity precisely when the sector-wise conditions hold—and s_1 predicts the monitored $\mathcal{C}_R$ across resolving measurements (Appendix H).

A scoping remark on vocabulary. When we speak of ETH-typical or "scrambled" eigenstates we mean the structural imprint of chaotic dynamics on *eigenstates*—the ETH matrix-element structure that spreads a code broadly and near-identically over the measured operator's eigenbasis—not dynamical operator growth during the monitored evolution: for eigenstate codes the unitary acts trivially (up to a relative phase) on the code subspace between measurements, so protection is a property of how the measurement resolves the code, exactly the two channels above. We neither measure dynamical chaos diagnostics (out-of-time-order correlators, level statistics) nor tune chaoticity independently of the code; the causal contrast this paper does establish is structural—hybridization and leak versus athermality—resolved by the purity scan of Sec. 3.6.

3 Results

3.1 Thermal codes win: the apparent scar advantage is a baseline artifact

Figure 1 compares $\mathcal{C}_R(p)$ for the scar code against the thermal ensemble. For both local-Z and collective (staggered-Z) monitoring the scar curve lies below the thermal-ensemble mean at every $p > 0$, and below the $\pm 1\sigma$ ensemble band at every $p > 0$ for the collective channel and for $p \leq 0.12$ for the local one (at $p = 0.14$, 0.16 both codes are nearly purified, $\mathcal{C}_R < 0.03$, and the residual gap is within the band). A single nearest-in-energy thermal pair sits at the 0th percentile of the ensemble ($\mathcal{C}_R = 0.53$ versus ensemble mean 0.87 at L=14, p=0.10, collective), which is why an unaveraged comparison spuriously favors the scar.

3.2 The disadvantage persists to the largest accessible size

Figure 2 shows $\Delta\mathcal{C}_R \equiv \mathcal{C}_R^{\text{scar}} - \langle\mathcal{C}_R^{\text{therm}}\rangle$ versus L at two rates. At $p = 0.02$, where most of the information survives ($\mathcal{C}_R^{\text{therm}} = 0.79$–$0.97$), the deficit is large and *grows* with size in the local channel: -0.16, -0.09, -0.16, -0.35, -0.38 for $L = 10$–18 (collective: -0.09 to -0.16). At the benchmark rate $p = 0.10$ it is negative for all $L = 10$–18 and both channels: -0.008 to -0.06 (local) and -0.25 to -0.32 (collective). The thermal-ensemble band excludes zero at every size for the collective channel, and at every size except $L = 12$ for the local one (there -0.008 ± 0.009, consistent with zero on the original grid). The band is a descriptive pair-to-pair spread, not a test on the mean difference, so the headline sign decisions are additionally backed by two audits: a hierarchical bootstrap

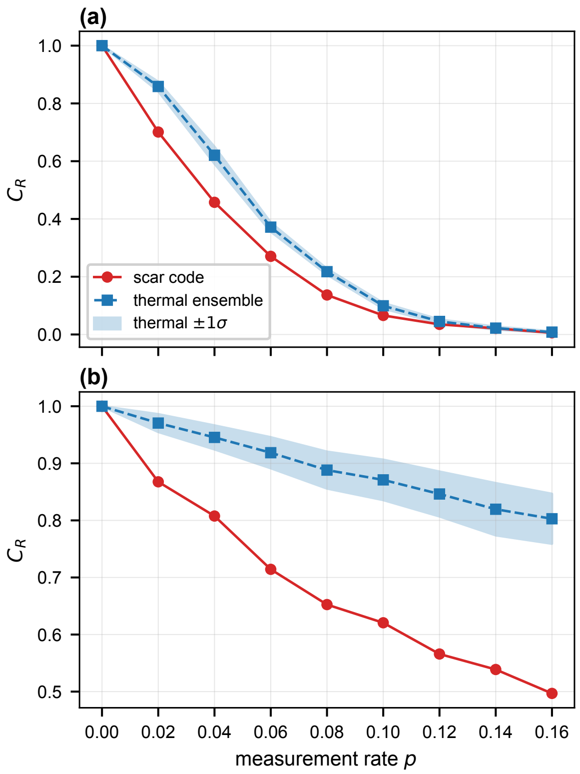

Figure 1: **Thermal codes protect information better than scar codes.** Reference-qubit coherent information $\mathcal{C}_R$ versus measurement rate p for the scar code (red) and the thermal ensemble (blue; band is $\pm 1\sigma$ over the ensemble) under (a) local-Z and (b) collective staggered-Z monitoring at $L = 14$. The scar lies below the thermal mean at every $p > 0$, and below the band except at the largest local rates, where both codes are nearly purified.

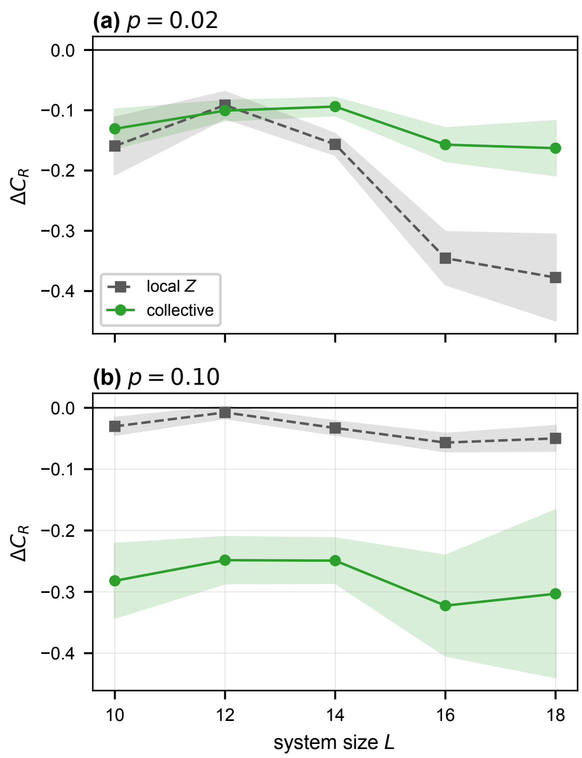

Figure 2: **The scar disadvantage $\Delta\mathcal{C}_R < 0$ at every size.** Scar-minus-thermal coherent information versus L for local (grey) and collective (green) monitoring at (a) $p = 0.02$, where most information survives and the local deficit grows with size, and (b) the benchmark $p = 0.10$; shading is the thermal-ensemble spread (12 pairs, 200 trajectories for $L = 10$–16; 6 pairs, 120 trajectories for $L = 18$). At $L = 16$ the overlap ranking selects the other member of the $2.64/2.74$ anticrossing doublet than at $L = 10$–14; the sign is partner-robust—the alternative partner gives -0.034 (local) and -0.150 (collective) against -0.066 and -0.306 at $p = 0.10$ (`scripts/l16_partner_check.py`).

that resamples thermal pairs and propagates per-code trajectory noise gives, at $p = 0.10$, 95% confidence intervals $[-0.056, -0.024]$ (local) and $[-0.317, -0.224]$ (collective) at $L = 14$ and $[-0.042, -0.012]$ for the marginal $L = 12$ local point; and a five-seed rerun of the trajectory noise finds every seed negative at all three headline points ($L = 12$ local -0.013 to -0.036, pooled -0.024, seed-to-seed s.d. 0.009; $L = 14$ local pooled -0.033; collective pooled -0.273) (`scripts/uncertainty_audit.py`, `nature_panel_fixes.py`). At the largest size $L = 18$ (constrained dimension 6765) the $p = 0.10$ gap is -0.05 (local) and -0.30 (collective), the collective gap essentially size-independent: the refutation is not a finite-size artifact, and at low rate it strengthens with size.

3.3 Even the maximal scar code loses

We emphasize that this single-qubit comparison is the most favorable one for the scar: the scar tower is only $(L+1)$-dimensional, so it can host at most $\log_2(L+1)$ logical qubits—a sub-extensive capacity, against the exponentially large thermal bulk. A scar subspace therefore cannot support an extensive code even before dynamics; the per-qubit comparison already settled here is its best case, and it loses. Within the overlap-identified (central, hybridized) rungs the deficit is also pair-robust: scanning every pair of positive-energy rungs from that set at $L = 10$ and 14, even the best-performing pair lies below the thermal-ensemble mean in both channels at $p = 0.10$ (at $L = 14$, best 0.081 versus thermal 0.099 local and 0.784 versus 0.865 collective; `scripts/referee_audit.py`). This robustness statement is deliberately scoped: the rung-resolved analysis of Sec. 3.5 shows that codes built at the *purer, outer* FSA rungs behave differently under local measurement at low rates, so the deficit is a property of the hybridized central rungs that the overlap criterion identifies—the experimentally accessed scar states—not of every conceivable tower encoding.

To confirm this directly, we encode the *maximal* scar code—all $k = \lfloor \log_2(L+1) \rfloor$ logical qubits, i.e. 2^k tower states maximally entangled with a k-qubit reference—and compare its protected-information density S_R/k against a same-dimension thermal code under local monitoring (Fig. 3). The scar loses at every rate: at $L = 14$ ($k = 3$), $\Delta(S_R/k) = -0.17, -0.14, -0.06$ at $p = 0.02, 0.04, 0.08$, with the thermal ensemble tightly determined ($\sigma \lesssim 0.01$), and the gap persists across $L = 10, 14, 18$. The gap is even larger under collective staggered-Z monitoring ($\Delta(S_R/k) = -0.32$ to -0.43 at $L = 14$), while under the fine-tuned Casimir J^2 the two codes are comparable (Δ up to $+0.034$, marginally scar-favorable; Appendix I)—the same exception seen in the single-qubit and spin-1 analyses. Even the largest code the scar subspace admits is thus a worse monitored memory than a generic thermal code of equal size under any generic measurement.

3.4 Mechanism: both channels resolve the scar code

The PXP scar realizes both channels of the criterion while thermal codes realize neither. Channel (ii) dominates under the collective staggered operator $M = \sum_i (-1)^i Z_i$—the staggered magnetization, order parameter of the emergent su(2) scar dynamics [32]—which carries a large logical leak $|\langle 0_L|M|1_L\rangle| = 1.73$ between the scar rungs (versus ≈ 0 for thermal codes) and coherently mixes them. We caution that the leak is one of the two channels, not a single-scalar predictor: the eigenstate sweep of Fig. 4(b) contains a leak-1.78 code with $\mathcal{C}_R = 0.79$, above the scar's 0.62, so a large leak *permits* but does not force poor protection—consistent with Appendix D's conclusion that no single scalar predicts $\mathcal{C}_R$; the quantity that does rank codes within a resolv-

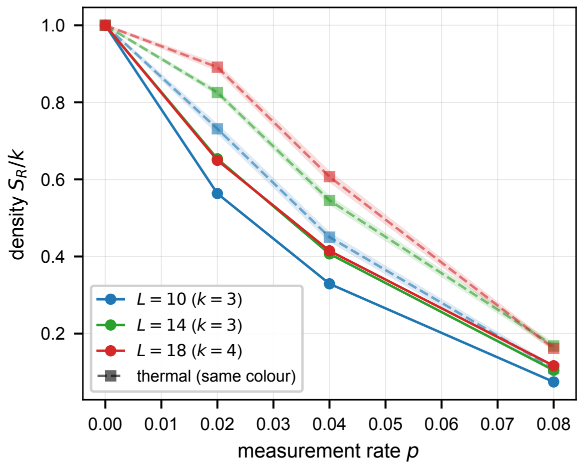

Figure 3: **The maximal scar code loses even as an extensive memory.** Protected-information density S_R/k for the maximal $k = \lfloor \log_2(L+1) \rfloor$-qubit scar code (solid) and a same-dimension thermal code (dashed, $\pm 1\sigma$ band) versus p under local monitoring, for $L = 10, 14, 18$. Scar lies below thermal throughout.

ing measurement is the sector-wise combination s_1 (Appendix H). Channel (i)—eigenvalue distinguishability through O-variance—is isolated by the Casimir measurement below. Accompanying both is a weak monotone trend [Fig. 4(a), Spearman $\rho = +0.41$]: protection tends to rise with the code's spread in the measurement basis, as ETH predicts, so that unstructured thermal codes are close to optimal.

The full (p, ℓ) dependence, with ℓ the measurement support size (block-sum of ℓ sites; ℓ=1 local, ℓ=L global), is negative throughout: the scar never wins (Fig. 8, Appendix E).

3.5 The decoherence-free-subspace rescue fails where scars are hybridized

A true DFS [26, 27] for the scar code would restore an advantage. The emergent Casimir $J^2 = \frac{1}{2}(H^+H^- + H^-H^+) + J_z^2$, built from the exact split $H = H^+ + H^-$ [32], gives the chiral-symmetric scar pair $(|{+E}\rangle, |{-E}\rangle)$ equal expectation and vanishing leak ($\Delta\langle J^2\rangle = 0.000$, leak $= 0.052$ at the canonical rung)—a nominal DFS. Yet under J^2 monitoring the scar code still loses, $\Delta\mathcal{C}_R = -0.18, -0.13, -0.09$ at $p = 0.04, 0.08, 0.12$ ($1.8\times$ the ensemble spread; an independent implementation—different energy window, ± 0.3 versus ± 0.5, different pairing rule, and different seeds—gives $-0.10, -0.08, -0.06$,

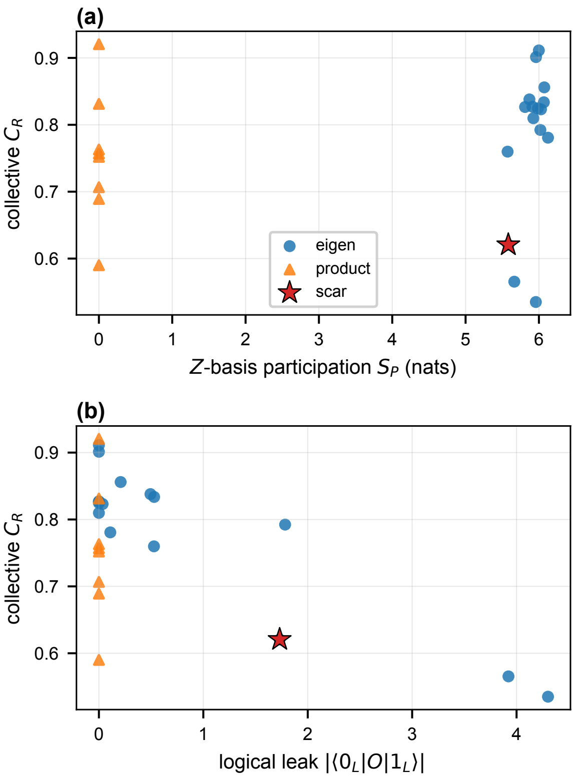

Figure 4: **What controls collective protection.** Collective $\mathcal{C}_R$ versus (a) Z-basis participation and (b) logical leak $|\langle 0|O|1\rangle|$ over an eigenstate sweep (blue), product states (orange), and the canonical scar (star). Higher participation raises $\mathcal{C}_R$; large leak suppresses it. The scar is a low-$\mathcal{C}_R$ outlier on both axes.

so the sign is robust while the magnitude depends on the ensemble construction, the window included; the J^2-channel window sensitivity is quantified in Appendix C).

Why it fails requires care, because the answer has two layers. The first is structural: the emergent J^2 at $L = 14$ has a dense, almost nondegenerate spectrum (987 eigenvalues with only six gaps exceeding 0.5), so a projective J^2 measurement at full spectral resolution resolves *any* two-dimensional code—the first shot collapses the scar pair outright, and its $\mathcal{C}_R$ reduces to the probability that no shot has yet occurred, *identically* at every rung (0.245, 0.033, 0.015 at $p = 0.04, 0.08, 0.12$, rung-independent to 13 digits). At full resolution no approximate-scar DFS is possible for spectral reasons alone, and the rung-to-rung variation of the raw deficit is entirely thermal-side statistics.

Can coarse-graining restore a DFS? Binning J^2 outcomes to the nearest $j(j+1)$ turns the measurement into a five-outcome probe under which a code *can* survive a shot; with thermal pools cleaned of all FSA-adjacent states (Appendix C) this binning rescues the purer chiral pairs ($\Delta\mathcal{C}_R = +0.18, +0.25, +0.20$ at $E = 5.34$) while the canonical rungs still lose ($-0.01, -0.11, -0.17$). We deliberately stress-tested this against alternative discretizations, and the result did *not* survive: with five equal-width bins, or with the $j(j+1)$ boundaries shifted by ± 1, the outcomes reorder completely—under one shift every chiral pair wins, the canonical one included ($+0.10$ to $+0.31$), and under the other the pure-rung rescue vanishes (-0.01 to -0.07) (`scripts/pure_rung_dfs2.py`, `nature_panel_fixes.py`). Coarse-Casimir survival is therefore controlled by where each code's $\langle J^2\rangle$ falls relative to the bin edges, and supports no purity—or variance—conclusion. What is construction-independent about the Casimir channel is only this: the chiral DFS candidates lose at full spectral resolution at every rung and under every ensemble construction tested, and no choice of binning makes the emergent, inexact J^2 of PXP a decoherence-free probe by design rather than by bin accident. For the statistics, at every energy-matchable rung the scar does retain excess Casimir variance over its cleaned thermal neighbors [Var ratio $6.9, 4.8, 4.4, 3.1$ at $E \simeq 2.7, 4.1, 5.3, 6.6$; Var J^2 itself falls $32.5 \to 8.6$ with FSA weight $0.53 \to 0.81$]. The exactly solvable model below supplies the clean end point where the algebra closes, the variance vanishes identically, and the rescue is exact—there, and only there, is the Casimir DFS a statement of structure rather than of discretization.

3.6 Protection tracks hybridization, not athermality

Collecting the rung-resolved data yields the single most consequential plot of this study (Fig. 5). Under generic local-Z monitoring at $p = 0.04$ the deficit rises *monotonically* with the rung's FSA weight, $\Delta\mathcal{C}_R = -0.069, +0.026, +0.069, +0.099$ at $w = 0.53, 0.66, 0.76, 0.81$, crossing from a scar deficit to a scar *advantage* near $w \simeq 0.65$; the ordering persists at $p = 0.08$ and flattens by $p = 0.12$. The crossover is driven by the scar side, not the baseline: along the same scan the thermal mean rises only from 0.65 to 0.73 while the scar rises from 0.58 to 0.83. (The coarse-grained

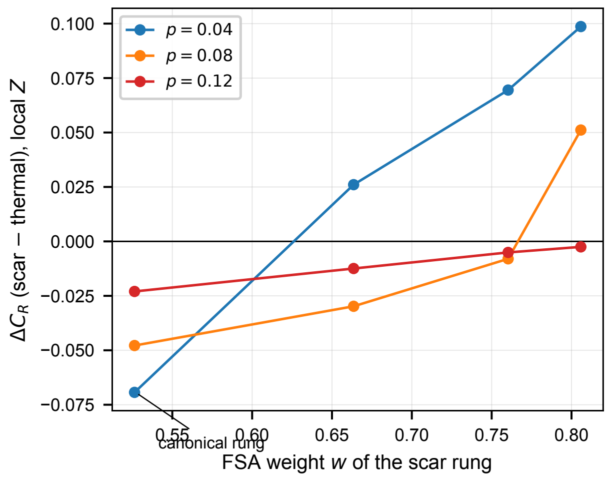

Figure 5: **Hybridization, not athermality, controls protection.** Scar-minus-thermal coherent information at each energy-matchable rung of the PXP tower versus the rung's FSA weight w (cleaned pools, $L = 14$; w increases from the canonical rung $E \simeq 2.74$ outward to $E \simeq 6.57$). (a) Local-Z monitoring: the deficit rises monotonically with purity and crosses to a scar advantage near $w \simeq 0.65$ at low rates. (b) Multiplet-binned Casimir: the same purity ordering, with bin geometry superimposed (Sec. 3.5).

Casimir is excluded from this plot: binning controls show its outcomes are set by bin geometry rather than purity, Sec. 3.5.) Since the purer rungs are the *more* athermal states, athermality itself is evidently not the liability—hybridization with the thermal bulk is, and it is maximal exactly at the canonical rungs that experiments access. Two caveats bound this reading: energy-matched partners at the outer rung energies are themselves increasingly atypical (Appendix C), and the collective channel is different in kind—its fragility comes from the su(2) generator's leak between adjacent rungs, an algebraic property of the natural scar encoding rather than of hybridization. The headline refutation therefore decomposes into two structural liabilities of the physically realized scar code: an algebraic leak that any SGA-aligned collective measurement exploits, and thermal hybridization that degrades local and Casimir protection.

3.7 A second model anchors the zero-variance end point

To test generality and to isolate this mechanism we repeat the analysis in the spin-1 Schecter–Iadecola magnet [33], whose scar tower $|S_n\rangle = (Q^+)^n |\Omega\rangle$, $Q^+ = \sum_j (-1)^j (S_j^+)^2$, is an *exact* su(2) multiplet. Two facts then hold to machine precision: the scars are exact eigenstates, and the Casimir J^2 is exactly constant on the tower with *zero* variance on every scar—against $\mathrm{Var}\, J^2 \simeq 32$ for PXP [Fig. 6(b)]. This flips the Casimir rescue [Fig. 6(a)]: the exact-su(2) scar code is now a genuine decoherence-free subspace—its two logical states are identical J^2 eigenstates, so a J^2 shot acts as the identity on the code and $\mathcal{C}_R^{\mathrm{scar}} = 1$ at every p *by construction* ($\Delta\mathcal{C}_R = +0.6$ to $+0.9$ against the thermal ensemble, $L = 6, 8$). The content of this exactly solvable limit is therefore not the protection itself but the contrast: the nominal DFS conditions that failed for PXP (equal expectation, zero leak) here come with zero variance, and protection is restored—anchoring, together with the hybridization-resolved rung scan of Sec. 3.5, the reading that what strips approximate scars of Casimir protection is their hybridization-induced variance (an exact end point plus a purity-resolved dependence, not an isolated causal test). This is parameter-independent: for $(h, D) = (0.5, 0.3)$ and $(1.5, 0.0)$ the Casimir variance again vanishes identically and $\mathcal{C}_R^{\mathrm{scar}} = 1$ under J^2. Under generic local S^z monitoring, by contrast, the exact-scar code shows a low-rate scar-favorable difference whose behavior is size-dependent, and we report it in full. At $L = 6$ (20 thermal pairs) it decays and changes sign within our window ($\Delta\mathcal{C}_R = +0.10, +0.06, -0.02$ at $p = 0.04, 0.08, 0.12$), as it does in two of three parameter sets; at $L = 8$ (20 pairs) it is larger and remains positive throughout ($\Delta\mathcal{C}_R = +0.20, +0.10, +0.08$, ensemble SEM 0.05–0.07). The same low-rate pattern appears in PXP at its purest rungs ($\Delta\mathcal{C}_R = +0.099$ at $p = 0.04$ for the $E \simeq \pm 6.6$ chiral pair with cleaned pools, decaying to 0 by $p = 0.12$; Sec. 3.5), with the caveat that energy-matched partners near the spectral edge are themselves atypical. Two sizes cannot settle whether this local advantage of *exact* scars survives at scale; what is settled is that it decays with rate and that no such advantage exists for the hybridized PXP scar code that experiments realize (Appendix G).

3.8 Relation to prior monitored-scar work

Our protocol maps to the continuous-time rate of Ref. [29] via $\gamma = p/dt$, so the range studied

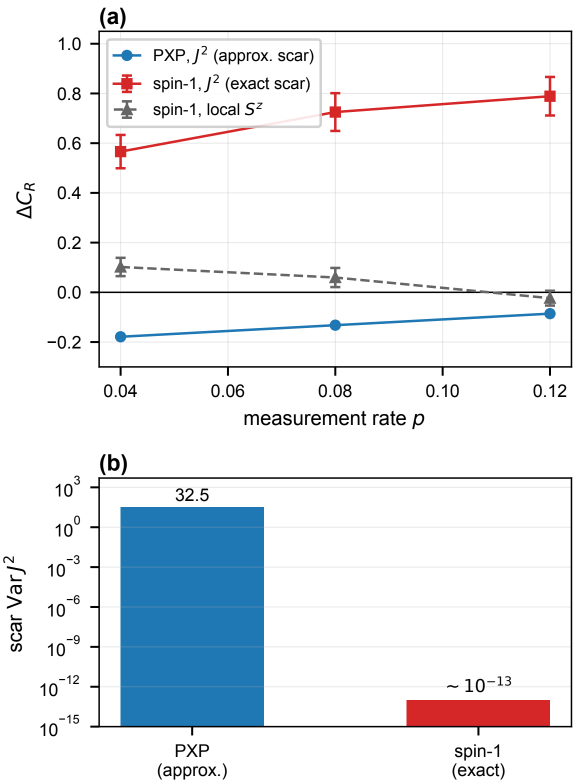

Figure 6: **Exact-su(2) scars evade the Casimir-variance obstruction.** Second model (spin-1 Schecter–Iadecola, $L = 6$, 20-pair thermal ensemble). (a) Scar–thermal coherent information $\Delta\mathcal{C}_R$ versus p (error bars are the ensemble SEM). Under J^2 monitoring PXP scars lose (blue) while exact spin-1 scars win perfectly (red); under generic local S^z the exact-scar and thermal codes are comparable at this size ($|\Delta\mathcal{C}_R| \leq 0.1$, grey; at $L = 8$ the low-rate difference is larger, $+0.20$ at $p = 0.04$, see text). (b) Scar Casimir variance, $\simeq 32$ (PXP, canonical rungs; 3.4–47.6 across the tower, always above energy-matched thermal neighbors) versus $\sim 10^{-13}$ (exact) on a log scale—the control parameter of the rescue.

here ($p \leq 0.16$, $\gamma \leq 0.27$) brackets their critical rate $\gamma_c \simeq 0.013 \pm 0.002$ from above. We emphasize that at our sizes and circuit depths the monitored-Néel entanglement density *decreases* with L at every p (Appendix F), i.e. we do not resolve the volume-law phase or its critical point, and we therefore make no quantitative comparison of critical rates; a minimum-spread estimate places the steepest-variation scale at $p \sim 0.05$ ($\gamma \sim 0.08$), consistent in order of magnitude but grid- and estimator-sensitive.

3.9 Experimental accessibility

The negative result is testable on the platform where PXP scars were first observed [21]. Programmable Rydberg-atom arrays natively realize the blockaded dynamics, local-Z monitoring is site-resolved projective readout interleaved at a tunable rate, and the sizes studied here ($L \leq 18$) are modest by current standards. Encoding in the scar subspace is accessible through the Néel revival manifold, and the retained coherence can be estimated by interferometric or randomized-measurement protocols. While preparing individual thermal eigenstates is impractical, thermal-like codes can be prepared operationally by pre-scrambling product states under the chaotic dynamics itself, and the scar code's excess fragility appears directly as a faster purification of the Néel-manifold encoding.

4 Conclusion

We quantified, as a monitored-channel coherent information, how well a logical qubit stored in the PXP quantum-many-body-scar subspace is protected against random measurement, benchmarked against an ensemble-averaged thermal code at matched energy—to our knowledge the first such head-to-head coding comparison of scar versus thermal subspaces under monitoring. For the single-qubit code, thermal encodings protect information better than the PXP scar encoding at every size ($L = 10$–18) and under every measurement model we test, by $\Delta\mathcal{C}_R \simeq -0.16$ (local, $p = 0.02$, where most of the information still survives) to -0.3 (collective, $p = 0.10$); for the maximal multi-qubit code the same holds under local and collective monitoring, with the fine-tuned Casimir comparable (Δ up to $+0.034$). The scar's emergent-su(2) order parameter supplies a logical leak of 1.73 that makes it especially fragile, and even a fine-tuned Casimir DFS fails for the chiral DFS candidates at full spectral resolution at every rung—scars retain excess Casimir variance over energy-matched thermal neighbors (ratio 3.1–6.9)—while coarse-grained Casimir outcomes prove to be bin-geometry artifacts supporting no rescue claim. This realizes the eigenstate-thermalization-to-approximate-error-correction correspondence [18–20] as an operational statement about monitored circuits, and sharpens it: what decides a code's fate is its structure with respect to the measurement (Sec. 2.2)—ETH-typical spread helps, a coupling generator and thermal hybridization hurt—not athermality per se, as the purity scan of Sec. 3.6 shows directly. Any program that would exploit the experimentally realized, hybridized scar towers for measurement-robust quantum memory should expect to lose to a generic chaotic subspace at the depths and sizes where the comparison is well posed (our claims are fixed-depth benchmarks, not asymptotic lifetimes); the relevant resource is the code's spread in the measurement basis, and structured non-thermal towers whose generators connect the code states are actively harmful. A second, exactly-solvable spin-1 model [33] whose tower closes the algebra exactly anchors the zero-variance end point—there the Casimir measurement protects the scar perfectly, by construction, while under generic local measurement its low-rate advantage decays with rate and its large-size fate is open (Sec. 3.7). Whether the PXP fragility survives generic non-projective noise channels and finite-temperature encodings [25, 34], and whether the purity-resolved local advantage of near-exact scars survives at scale, are the natural next questions.

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Author contributions

H.W. and T.T. conceived the study. H.W. developed the methodology, implemented the software, performed all computations and analysis, produced the figures, and wrote the manuscript.

Data and code availability

The simulation package, drivers, and the data underlying all figures are openly available at doi:10.5281/zenodo.21642056 (source: `https://github.com/deeptell-inc/chaosec`), enabling one-command reproduction of every result reported here.

A Constrained basis, PXP spectrum, and scar identification

The PXP chain is diagonalized in the Rydberg-blockaded basis of bit strings with no two adjacent excitations (dimension $F(L{+}2)$, Fibonacci; open boundaries). We benchmark the implementation against the canonical $\mathbb{Z}_2$-Néel revival: for $L = 12$ the return fidelity $|\langle \mathbb{Z}_2|e^{-iHt}|\mathbb{Z}_2\rangle|^2$ peaks at $t \simeq 4.70$ with height 0.743, reproducing the literature. The $L{+}1$ scar-tower eigenstates are selected as the states with anomalously large $|\langle \mathbb{Z}_2|E_k\rangle|^2$. The forward-scattering (FSA) tower built from $|\mathbb{Z}_2\rangle$ with the $H = H^+ + H^-$ split spans an approximately equally spaced ladder out to the spectral edge ($E_{\mathrm{FSA}} \simeq 0, \pm1.34, \pm2.67, \pm3.98, \pm5.25, \pm6.47, \pm7.63, \pm8.68$ at $L = 14$). Two caveats about the overlap-based selection deserve disclosure. First, near the center of the spectrum the FSA modes hybridize strongly with the thermal bulk—the exact eigenstate closest to each FSA rung carries FSA weight $0.97, 0.88, 0.81, 0.76, 0.66, 0.53, 0.27$ from the edge rung inward (`scripts/fsa_purity_audit.py`)—so at $L = 14$ the top-($L{+}1$) overlap selection concentrates on the central rungs and splits across near-degenerate doublets at anticrossings (e.g. 2.627/2.737 around the 2.67 rung) rather than returning one state per rung. The single-qubit code of the main text uses the eigenstates of locally maximal overlap at the $E \simeq 1.33$ and 2.74 rungs—the conventional, experimentally relevant scar states, but also the most hybridized rungs of the tower; the dependence of our conclusions on this choice is audited in Sec. 3 and Appendix I. Second, PXP possesses an exponentially large $E = 0$ degeneracy ($8, 13, 21$ zero modes at $L = 10, 12, 14$), within which eigenvectors are basis-arbitrary; single-qubit scar and thermal codes are therefore built from positive-energy rungs ($|E| > 0.8$) to avoid this manifold.

B Monitored-trajectory algorithm

A logical qubit maximally entangled with a reference R is stored as $\phi \in \mathbb{C}^{d\times 2}$, with column r holding $\langle r_R|\Psi\rangle$ (an unnormalized system vector of dimension d). Each step applies the dense propagator $U = e^{-iH\,dt}$ to both columns and then, with probability p, performs a projective measurement: for a diagonal operator with values $\{v\}$ we Born-sample an eigenvalue from weights $\sum_r \sum_{q:v_q=v} |\phi_{qr}|^2$ and project. The reference density matrix is $\rho_R = \phi^\dagger \phi$ (a 2×2 Hermitian matrix of unit trace), and $S_R = -\mathrm{Tr}\, \rho_R \log_2 \rho_R$. We report $\mathcal{C}_R$ as the time average of S_R over the final $\lfloor N/4 \rfloor = 2$ of the $N = 10$ recorded points of a 40-step run (S_R recorded every 4 steps; the two records cover $t \in [21.6, 24]$, i.e. 20% of the records and the last 10% of the evolution window), averaged over 120–400 Born-sampled trajectories with fixed seeds; error bars are the standard error over trajectories. As a consistency check, $p = 0$ yields $\mathcal{C}_R = 1$ identically (a unitary on the system cannot reduce the reference entanglement), and $\mathcal{C}_R \to 0$ at large p.

$\mathcal{C}_R$ so defined decays with circuit depth—a finite monitored system purifies—so it is a fixed-depth quantity, and all comparisons in this paper are at matched depth. Rerunning the L=14, p=0.10 headline point at depths 40, 80, and 160 steps gives scar-minus-thermal gaps of -0.270, -0.338, -0.325 in the collective channel: there the deficit is depth-stable while the absolute values decay. In the local channel at $p = 0.10$ both codes are essentially purified beyond depth 80 (-0.040, -0.0015, $+7 \times 10^{-5}$; the last value is sign-reversed but both codes sit at $\mathcal{C}_R \sim 10^{-4}$, so it carries no information), so depth-robustness of the local ordering must be assessed at a lower rate, where information survives the added depth: at $p = 0.02$ the local gap is -0.144, -0.151, -0.127 at depths 40, 80, 160 (scar 0.72/0.57/0.38 against thermal 0.87/0.73/0.51)—depth-stable like the collective channel (`scripts/referee_audit.py`, `uncertainty_audit.py`).

For a general (non-diagonal) Hermitian measurement operator O (e.g. the su(2) Casimir J^2) we precompute its eigenbasis once, rotate ϕ into it, project onto a Born-sampled eigenvalue group, and rotate back.

C Thermal-ensemble baseline and the single-pair artifact

The thermal baseline must be averaged over many energy-matched thermal pairs. The naive choice—the single non-scar eigenstate nearest each scar-rung energy—is systematically biased toward atypical, near-scar states with anomalously low $\mathcal{C}_R$. At $L = 14$, $p = 0.10$, collective monitoring, that single pair gives $\mathcal{C}_R = 0.53$, far below every member of the shipped 12-pair ensemble (mean 0.87 ± 0.04; the pair-resolved audit of `scripts/uncertainty_audit.py` gives minimum 0.80, maximum 0.93). Averaging over the ensemble reverses the sign of $\Delta\mathcal{C}_R$ and is essential to the conclusion; all results in the main text use the ensemble baseline.

C.1 Energy-matching tolerance

The ensemble is selected within a finite energy window, not at exactly the scar-rung energies: each member is a generic (non-tower, $|\langle \mathbb{Z}_2 | E \rangle|^2 < 0.02$) eigenstate within ± 0.5 of one of the two targets. At $L = 14$ the realized offsets are 0.22 on average and 0.47 at most, so the matching is approximate. Three checks, all at the headline point (L=14, p=0.10, 12 pairs, 250 trajectories, identical seeds), show this tolerance does not produce the scar deficit. *(i)* Tightening the window from ± 0.5 to ± 0.25 and ± 0.10 (mean offset $0.22 \to 0.13 \to 0.06$) leaves the deficit essentially unchanged: $\Delta\mathcal{C}_R = -0.033$, -0.034, -0.031 (local) and -0.25, -0.23, -0.21 (collective); at the tightest matching the gaps remain ≈ 2.4 and ≈ 2.8 ensemble standard deviations below zero. (At the tightest window the matching pool is smaller than the ensemble, so some eigenstates recur across pairs; the headline ± 0.5 ensembles use 24 distinct states with no reuse.) *(ii)* Within the ensemble the per-pair $\mathcal{C}_R$ is uncorrelated with the pair's mean absolute offset [Spearman $\rho = +0.05$ ($p = 0.88$, local) and 0.00 ($p = 1.00$, collective) at ± 0.5], so the thermal band reflects state-to-state typicality, not residual energy dependence. For completeness, the per-pair $\mathcal{C}_R$ does correlate with the *signed* total offset in the collective channel [$\rho = +0.56$ ($p = 0.06$) at ± 0.5; $+0.67$ ($p = 0.02$) at ± 0.10; no consistent sign in the local channel, $n = 12$ pairs]: thermal pairs sitting slightly above the rung energies protect marginally better. The window-tightening in *(i)* bounds the size of this effect—in the zero-offset limit the collective thermal mean extrapolates to ≈ 0.83, still 0.21 above the scar. *(iii)* That weak drift with the window (collective thermal mean $0.87 \to 0.83$) slightly *shrinks* the gap, the opposite of an energy-mismatch artifact inflating it. The spin-1 and extensive-code analyses use analogous windows (± 0.8 and ± 0.5). The audit is reproduced by `scripts/energy_window_robustness.py`.

C.2 The Casimir channel: window sensitivity and pool hygiene

The window audit above covers the local and collective channels; the fine-resolution J^2 channel behaves differently and we disclose it separately. There the scar-side $\mathcal{C}_R$ is pure counting statistics (Sec. 3.5), so the deficit magnitude is set entirely by the thermal side and inherits its sampling sensitivity: at the canonical rung with a fixed seed, windows ± 0.3 and ± 0.5 give $\Delta\mathcal{C}_R = -0.050, -0.035, -0.023$ versus $-0.161, -0.135, -0.088$ at $p = 0.04, 0.08, 0.12$, and the two independent implementations quoted in Sec. 3.5 give -0.18 and -0.10 at $p = 0.04$. The sign is negative in all four constructions; magnitudes should be read with 3–4× construction sensitivity in mind (`scripts/pure_rung_dfs2.py`).

The Casimir channel is, moreover, *encoding-dependent* in a way the generic channels are not. All DFS-oriented results in Sec. 3.5 use the chiral pair $(|{+E}\rangle, |{-E}\rangle)$, the nominal DFS candidate, which loses at the canonical rungs under every construction tested. The *adjacent-rung* code of the main comparison ($E \simeq 1.33, 2.74$) behaves oppositely under the fine-resolution Casimir: because its two rungs share the emergent multiplet structure, their J^2 outcome distributions overlap strongly [BC $= 0.958$, Fig. 7] while the leak stays small, so the shot resolves thermal codes more strongly than the scar code and the scar *wins*: $\Delta\mathcal{C}_R = +0.19$, $+0.15$, $+0.13$ at $p = 0.04, 0.08, 0.12$ with hierarchical 95% CIs $[+0.11, +0.27]$, $[+0.09, +0.21]$, $[+0.08, +0.18]$ (under the multiplet-binned Casimir the same code is comparable, CIs straddling zero). This is exactly what the two-channel criterion predicts, and it is why the abstract's ordering claim is scoped to *generic* measurements: a probe fine-tuned to the scars' shared algebraic structure can invert it (`scripts/uncertainty_audit.py --part-c`).

Pool hygiene for the rung-resolved analysis: the standard exclusion list is the top-$(L{+}1)$ overlap mask, which does not contain the outer FSA rungs, so a first-pass rung scan inadvertently admitted the scar eigenstates themselves into their own "thermal" pools at $E = 6.57$ and 7.71. All rung-resolved numbers quoted in this paper come from the corrected scan, which explicitly excludes the eigenstate nearest to every FSA rung and deduplicates chiral partners in variance statistics (20 distinct pool states per rung; the $E = 7.71$ rung retains only 4 distinct partners in the ± 0.5 window and is therefore excluded from the dynamical comparison).

D Static Knill–Laflamme diagnostics

For a code $\{|0_L\rangle, |1_L\rangle\}$ and error set $\{E_a\}$ we quantify correctability through the logical leak $u_a = \langle 0_L|E_a|1_L\rangle$ and distinguishability $d_a = \langle 0_L|E_a|0_L\rangle - \langle 1_L|E_a|1_L\rangle$. Under the local-$Z$ set, energy-matched scar codes have a larger root-mean-square violation than thermal codes (ratio 1.4–2.4 for $L = 10$–14), consistent with ETH-based approximate error correction. Under the collective staggered operator $M = \sum_i (-1)^i Z_i$ the scar rungs, being connected by the emergent su(2) generator, carry a large leak $|\langle 0_L|M|1_L\rangle| = 1.73$ (versus ≈ 0 for thermal codes), which drives the strong scar penalty under SGA-aligned monitoring.

The two channels of the main-text criterion are diagnostically distinct. The diagonal outcome-distribution overlap alone—the Bhattacharyya coefficient BC $= \sum_o \sqrt{P_0(o)P_1(o)}$ of the two logical states under the collective M—does *not* predict $\mathcal{C}_R$ (Fig. 7): the scar has a high overlap BC $= 0.96$ yet is poorly protected, because its M-fragility comes from channel (ii), the off-diagonal leak, which BC cannot see. Both channels are therefore required; no single scalar (participation, variance, or leak alone) predicts protection across all codes.

E Measurement-support phase diagram

Figure 8 maps $\Delta\mathcal{C}_R(p, \ell)$ where ℓ is the support size of a block-sum measurement $\sum_{i\in W} Z_i$ over a random contiguous window W of ℓ sites, measured with probability p *per step* (one random window per event). Note that $\ell{=}1$ under this block protocol is therefore *not* identical to the per-site local protocol of the main text, which attempts each of the L sites independently per step. Against the ensemble thermal baseline the map is negative at every grid point—the scar code never wins—and the disadvantage

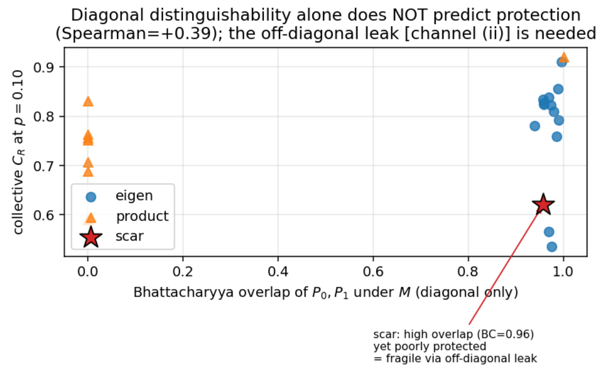

Figure 7: Collective $\mathcal{C}_R$ versus the diagonal Bhattacharyya overlap of the two logical states under M ($L = 14$). The scar (star) has near-unit overlap yet low $\mathcal{C}_R$: its fragility is off-diagonal [channel (ii)], invisible to BC.

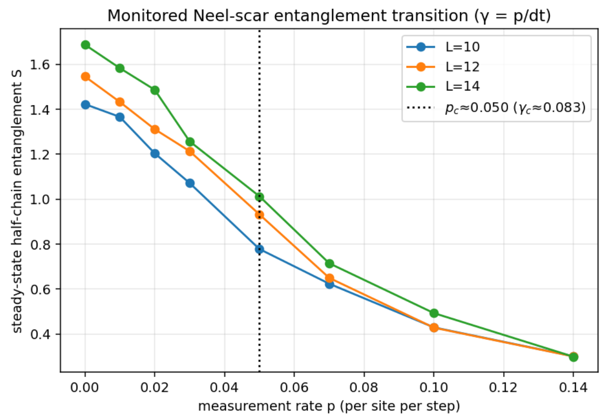

Figure 9: Half-chain entanglement of the monitored Néel state versus p for $L = 10, 12, 14$. At these sizes and depths the density decreases with L at every p—the curves do not cross—so the marked $p \simeq 0.05$ is the scale of steepest variation, not a critical point.

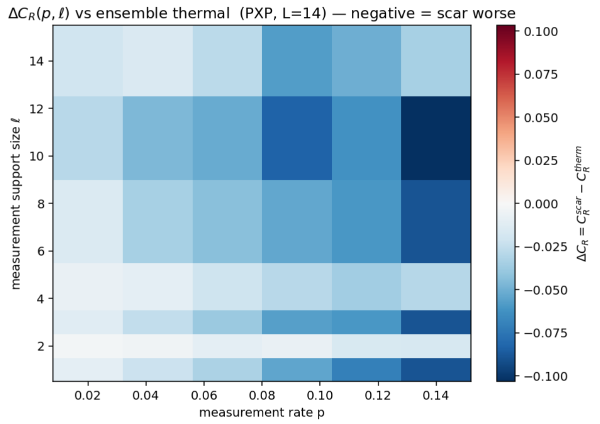

Figure 8: $\Delta\mathcal{C}_R(p, \ell)$ against the ensemble thermal baseline (PXP, $L = 14$). Blue is $\Delta\mathcal{C}_R < 0$ (scar worse) everywhere.

deepens with p; its ℓ-dependence, however, is *non-monotone*, with even-ℓ supports systematically milder than adjacent odd ones (at $p = 0.14$: $-0.089, -0.018, -0.089, -0.030, -0.090, -0.103, -0.034$ for $\ell = 1, 2, 3, 4, 7, 11, 14$), reflecting how a contiguous block couples to the staggered scar structure. (The staggered SGA operator of the main text produces a larger penalty than the uniform-type block sum because the latter has vanishing logical leak on the scar rungs.)

F Calibration of the measurement rate

To connect our discrete per-step rate to the continuous-time rate of Ref. [29] we use $\gamma = p/dt$ with $dt = 0.6$. An honest caveat applies: at the sizes and depths accessible here the monitored-Néel entanglement density $S/(L/2)$ decreases with L at *every* p (at $p = 0$: 0.285, 0.258, 0.241 for $L = 10, 12, 14$), so the curves in Fig. 9 do not cross and no volume-law phase—hence no critical point—is resolved. The minimum of the inter-size spread of $S/(L/2)$ sits at $p \simeq 0.05$ ($\gamma \simeq 0.083$), but nearby rates give almost the same spread (0.0051 at $p = 0.05$ versus 0.0071–0.0072 at $p = 0.10$–0.14), so we treat this only as the scale of steepest variation. It is of the same order as the reported $\gamma_c \simeq 0.013 \pm 0.002$; we do not attempt a quantitative comparison of critical rates.

G Spin-1 Schecter–Iadecola model

The second model is the spin-1 chain [33]

$$H = J \sum_i (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y) + h \sum_i S_i^z + D \sum_i (S_i^z)^2, \tag{3}$$

with $J{=}1$, $h{=}1$, $D{=}0.1$. Its exact scar tower is $|S_n\rangle = (Q^+)^n |\Omega\rangle$, $Q^+ = \sum_j (-1)^j (S_j^+)^2$, $|\Omega\rangle = |m{=}{-1}\rangle^{\otimes L}$. The site operators $\{|m{=}{-1}\rangle, |m{=}{+1}\rangle\}$ realize a pseudospin-$\frac{1}{2}$ (with $|m{=}0\rangle$ outside), and the tower is the fully symmetric Dicke multiplet of total pseudospin $J = L/2$. With the properly normalized generators $J^+ = Q^+/2$, $J^- = (J^+)^\dagger$, $J^z = \frac{1}{2}\sum_i S_i^z$ one has $[J^z, J^\pm] = \pm J^\pm$, $[J^+, J^-] = 2J^z$, so the Casimir $J^2 = (J^z)^2 + \frac{1}{2}(J^+J^- + J^-J^+)$ equals $J(J{+}1) = \frac{L}{2}(\frac{L}{2}{+}1)$.

We verify numerically that $|S_n\rangle$ are exact eigenstates (residual $\|H\,|S_n\rangle - E_n\,|S_n\rangle\| \sim 10^{-16}$), equally spaced by 2, and that J^2 is constant on the tower with variance $\sim 10^{-13}$ on every scar—versus $\operatorname{Var} J^2 \simeq 32$ for PXP scars. Consequently a two-rung scar code has identical J^2 outcome distributions and is an exact decoherence-free subspace of a J^2 measurement, which is why the rescue succeeds here and fails for PXP ($\mathcal{C}_R^{\rm scar} = 1$, $\Delta\mathcal{C}_R = +0.6$ to $+0.9$ at $L = 6, 8$). Under generic local S^z monitoring the exact-scar code shows a low-rate advantage whose size dependence we report in full. At $L = 6$ (20 pairs) $\Delta\mathcal{C}_R = +0.10, +0.06, -0.02$ at $p = 0.04, 0.08, 0.12$ (ensemble SEM ~ 0.04): scar-favorable at the lowest rate (about 2σ), decaying and reversing sign with p, a pattern repeated in two of the three parameter sets [$(h, D) = (1.5, 0.0)$: $+0.11, +0.03, -0.02$; $(0.5, 0.3)$: $+0.05, -0.03, -0.05$]. At $L = 8$ (20 pairs) the difference is larger and stays positive across our window: $\Delta\mathcal{C}_R = +0.20, +0.10, +0.08$ (ensemble SEM 0.05–0.07, so $\approx 3\sigma$ at $p = 0.04$). With two sizes we cannot settle the large-L fate of this local-measurement advantage; what the data establish is that it decays with rate, and that the unconditional scar-versus-thermal reversal seen for PXP does not occur for exact scars.

H A single-shot resolvability predictor

The two channels of the main-text criterion combine into one exact, simulation-free scalar: the reference entropy that survives a *single* projective O-measurement of the freshly encoded pair,

$$s_1(O) = \sum_g q_g\, S(\rho_R^{(g)}), \qquad [\rho_R^{(g)}]_{ab} = \frac{\langle b_L|\Pi_g|a_L\rangle}{2q_g}, \tag{4}$$

with Π_g the spectral projectors of O and $q_g = \frac{1}{2}(\langle 0_L|\Pi_g|0_L\rangle + \langle 1_L|\Pi_g|1_L\rangle)$ the Born weights. By construction $s_1 = 1$ iff a single shot leaves the qubit perfectly recoverable—the Knill–Laflamme conditions for the projector set $\{\Pi_g\}$, of which a common-eigenvalue eigenspace (a true DFS) is the simplest realization—and $s_1 = 0$ iff one shot fully reads the qubit out; eigenvalue distinguishability [channel (i)] and non-invariance [channel (ii)] both enter automatically.

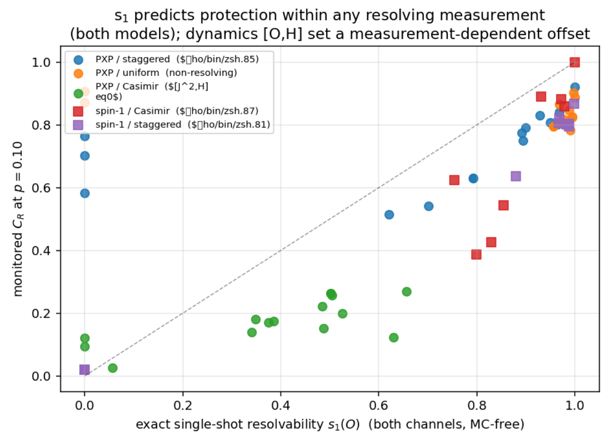

Figure 10: Monitored $\mathcal{C}_R$ ($p = 0.10$) versus the exact single-shot resolvability $s_1(O)$ for 65 code–measurement pairs across both models. Within each resolving measurement the relation is tight; the non-resolving (uniform) and non-commuting (PXP Casimir) families bound the predictor's scope.

Across 65 code–measurement pairs spanning both models and five global operators (Fig. 10), s_1 strongly predicts the monitored $\mathcal{C}_R$ *within each resolving measurement*: Spearman $\rho = +0.84$ (PXP, staggered), $+0.87$ and $+0.79$ (spin-1, Casimir and staggered). Two limiting behaviors delimit its scope and are themselves instructive: for a non-resolving operator (PXP uniform, $s_1 \approx 1$ for all codes) there is no signal left to predict ($\rho = -0.09$, $p = 0.75$), and for an operator that does not commute with the Hamiltonian (the PXP Casimir, $[J^2, H] \neq 0$) the ranking survives ($\rho = +0.69$, $p = 0.004$) but the dynamics between shots dress the resolvability and shift $\mathcal{C}_R$ systematically below the single-shot estimate. We therefore present s_1 as an operational design quantity—an exact, Monte-Carlo-free screen for candidate code/measurement pairs—rather than a universal collapse variable.

I Extensive code under collective and Casimir monitoring

The maximal $k = \lfloor \log_2(L{+}1) \rfloor$-qubit scar code of the main text is compared to a same-dimension thermal code under all three PXP measurement models in Table 1 ($L = 14$, $k = 3$, protected density S_R/k, 8-draw thermal ensemble). The pattern matches the single-qubit analysis: the scar loses under local and (strongly) collective monitoring and is only comparable (marginally scar-favorable, up to +0.034) under the fine-tuned Casimir—the one measurement model in which the thermal code does not win. The local run used the grid $p = 0, 0.02, 0.04, 0.08$ of Fig. 3 (its $p = 0.02$ gap is -0.17); the collective and Casimir runs used $p = 0, 0.04, 0.08, 0.12$, hence the ragged table.

Two selection details deserve disclosure. The 2^k code states are the tower states of largest $\mathbb{Z}_2$ overlap, which at $L = 14$ comprise six $\pm E$ rung partners, the $E = 4.06$ rung, and one $E \simeq 0$ tower state; because $\pm 1.329/\pm 1.332$ are the two members of one anticrossing doublet, these eight states span only three distinct FSA rungs plus the zero mode, so "2^k tower states" means overlap-ranked eigenstates, not 2^k distinct rungs. The positive-rung restriction of the single-qubit analysis (Appendix A) is not imposed, since $2^k = 8$ exceeds the six positive rungs available. Excluding the zero mode entirely (selecting the top-8 among tower states with $|E| > 0.8$) changes nothing: the local densities move from 0.674/0.393/0.101 to 0.661/0.381/0.104 at $p = 0.02/0.04/0.08$ (both from the audit run, 150 trajectories, seed 1), far below the thermal 0.83/0.55/0.17 of the main run (200 trajectories, seed 10; the audit's default-selection scar densities agree with the main run's 0.654/0.407/0.105 to within trajectory noise) (`scripts/referee_audit.py`).

Table 1: Extensive-code density gap $\Delta(S_R/k)$ = scar − thermal ($L = 14$, $k = 3$).

measurement	p=0.04	p=0.08	p=0.12
local S^z	−0.14	−0.06	—
collective M	−0.32	−0.40	−0.43
Casimir J^2	+0.02	+0.03	+0.03

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